Annexure 2

PROBLEM SOLVING TECHNIQUES LAB

Course Code: CSE 624 **Credit Units: 02**
Total Hours: 60

Course Objective:
To write the programs for solving problems using the concept of C,C++ and data structures and develop knowledge of basic data structures for storage and retrieval of ordered or unordered data.

Software Requirements: Turbo C++ compiler

Course Contents :
Lab Experiments are based on the course Problem Solving Techniques (CSE 604)

List of experiments/demonstrations:

(A) Programming in C : (20 Hours)

1. Write a simple program based on operators (pre, post increment , bitwise and , or , etc.).
2. Write a simple program based on conversions (from int to float & float to int)
3. Write a program for find the max and min from the three numbers.
4. Write the program for the simple, compound interest.
5. Write program for students marks grading.
6. C program to check whether a given number is odd or even.
7. C program to Add digits of input number.
8. C program to Factorial of a given number.
9. C program to swap two numbers without using third variable.
10. C program to check whether a given year is leap year or not.
11. C program to check whether a given number Palindrome Number or not.
12. C programs to print different patterns.
13. Program for the following using switch statement:
    Menu:-
    (a) Sum of two numbers
    (b) Negative or Positive Number
    (c) Simple Interest
    (d) Area of Circle
    (e) Exit
14. C program to check whether a given number Prime Number or not.
15. C program to check whether a given number Armstrong Number or not.
16. C program to print Fibonacci series up to given term.
17. C program to find out sum of 10 numbers by using array.
18. C program to reverses of one array elements into another.
19. C program to find out maximum and minimum number in an array.
20. Write a C program that uses functions to perform the following:
    (a) Addition of Two Matrices
    (b) Multiplication of Two Matrices
    (c) Transpose of a matrix
21. C program to Factorial of a given number by using user define function.
22. Write a program for display values reverse order from array using pointer.
23. Write a program through pointer variable to sum of n elements from array .
24. Write a C program which copies one file to another.

(B) Object Oriented Programming in C++ : (20 Hours)

1. Write a program that show the concept of class and object and having function for addition,subtraction,multiplication and division of two number.

2. Program that show the concept of inline function.
3. Program that show the concept of friend function.
4. Program that show the concept of all types of constructor and destructor.
5. Program that show the concept of local class and global class.
6. Program that show the concept of local object and global object.
7. Program that show the concept of static class data and static member function.
8. Program that show the concept of constant member data and function .
9. Program that show the concept of dynamic memory allocation.
10. Program that show the concept of multiple inheritance.
11. Program that show the concept of multilevel inheritance.
12. Program that show the concept of function overloading.
13. Program that show the concept of function overriding.
14. Program that illustrates the order of execution of constructors and destructors when new class is derived from more than one base class.
15. Program that show the concept of operator overloading(overload ++ operator) .
16. Program that overload +,- for addition and subtraction of two complex number.
17. Program that show the concept of this pointer.
18. Program that illustrates how run time polymorphism is achieved using virtual functions.
19. Program that illustrates the role of virtual base class in building class hierarchy.
20. Program that illustrates the role of abstract class in building class hierarchy.

(C) Data Structure : (30 Hours)

1. Write a C/C++ program that uses functions to perform the following: i) Create a singly linked list of integers. ii) Delete a given integer from the above linked list. iii) Display the contents of the above list after deletion.
2. Write a C/C++ program that uses functions to perform the following: i) Create a doubly linked list of integers. ii) Delete a given integer from the above doubly linked list. iii) Display the contents of the above list after deletion.
3. Write a C/C++ program that implement the concept of Stack using array/link list.
4. Write a C/C++ program that implement the concept of Queue using array/link list..
5. Write a C/C++ program that implement the concept of Circular Queue.
6. Write a C/C++ program that implement the solution of Tower of Hanoi problem.
7. Write a C/C++ program that uses stack operations to convert a given infix expression into its postfix Equivalent.
8. Write a C/C++ program that uses functions to perform the following: i) Create a binary search tree of characters. ii) Traverse the above Binary search tree recursively in postorder.
9. Write a C/C++ program that uses functions to perform the following: i) Create a binary search tree of integers. ii) Traverse the above Binary search tree non recursively in order.
10. Write C/C++ programs for implementing the following sorting methods to arrange a list of integers in ascending order: i) Insertion sort ii) Bubble Sort iii) Insertion Sort iv) Quick Sort v) Merge sort vi) Counting Sort etc.
11. Write C/C++ programs for implementing the following graph traversal algorithms:
    (i)Depth first traversal (ii)Breadth first traversal

Course outcomes:

Able to write the program using different data structures.

- Able to implement linked list data structure to solve various problems.
- Able to apply various data structure such as stacks, queues, trees and graphs to solve various computing problems using C/C++ -programming language.
- To apply concepts and techniques for implementation.

Examination Scheme:

IA	EE